Newtonian Taylor–Culick retraction at arbitrary Ohnesorge number

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August 29, 2026

Abstract

A freely suspended planar liquid sheet, initially at rest, retracts under its own surface tension. The collected mass gathers in a growing rim. Taylor [1] and Culick [2] predicted a terminal speed $V_{TC} = \sqrt{2\sigma/(\rho h_0)}$ from a global momentum budget, independent of viscosity. Appendix A of Sanjay *et al.* [3] treated the Newtonian problem at a single film Ohnesorge number with a two-phase volume-of-fluid method. Here we sweep Ohnesorge number with a single-phase sharp-interface finite-element method in a frame co-moving with the rim. The recovered tip speed approaches V_{TC} at every $\mathrm{Oh_{SB}}$ in the production range 0.035 to 1. In a middle band $0.14724 \le \mathrm{Oh_{SB}} \le 1$ the deficit follows $1 - u^* = (1/(2\sqrt{\pi}))\,\sqrt{\mathrm{Oh_{SB}}/t^*}$ to within 1%. That coefficient is not universal on the evidence available. It departs at both ends of the Ohnesorge range. The strongly inertial end, $\mathrm{Oh_{SB}} < 0.1$, is closed against V_{TC} and against an independent Basilisk calculation rather than against lubrication curves. The one-dimensional long-wave model and the two-dimensional Navier–Stokes solution cannot agree across Oh in principle, because their deficits decay as $(t^*)^{-2}$ and $(t^*)^{-1/2}$ respectively.

1 Problem

A semi-infinite planar liquid sheet of uniform thickness h_0, initially at rest, retracts under its own surface tension σ. Surface tension pulls the free edge back. The collected mass accumulates in a growing rim. There is no surrounding gas. The interface is a true free surface with a stress condition on it, not an interface between two resolved fluids.

Taylor [1] and Culick [2] obtained the terminal retraction velocity by balancing the capillary pull against the rate of momentum gain of the rim,

$$2\sigma = \rho h_0 V_{TC}^2 \qquad \Rightarrow \qquad V_{TC} = \sqrt{2\sigma/(\rho h_0)}. \tag{1}$$

The velocity is set by a global momentum budget, so it is independent of viscosity. Viscosity controls only how the sheet approaches V_{TC}, and how long it takes.

Figure 1 shows the computational setup. This note is intended as an extension of Appendix A of Sanjay *et al.* [3], which treated the classical retraction at a single film Ohnesorge number ($\mathrm{Oh}_f = 0.05$) with Basilisk [4]. The numerical method here is different in kind. A single-phase, sharp-interface finite-element method in a frame co-moving with the rim, built on pyoomph [5] and oomph-lib [6]. The inertial planar comparison against Basilisk at $\mathrm{Oh_{SB}} < 0.1$ is closed and is reported in the companion inertial note. The one-dimensional Trouton/Ahsan work is the viscous-limit note.

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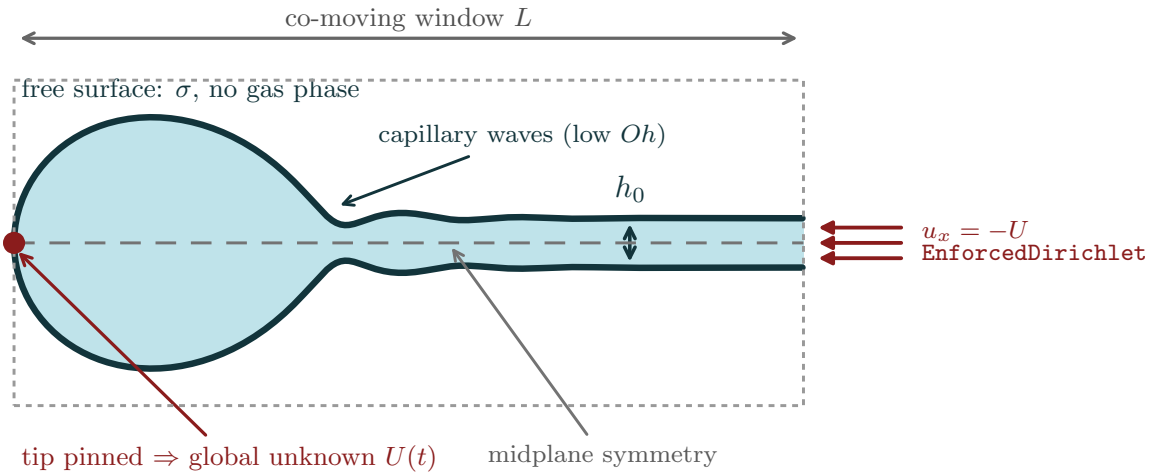


Figure 1: Planar sheet of thickness h_0 retracting to the left. The computation is performed in a frame co-moving with the rim. The tip is pinned at the origin and the frame velocity $U(t)$ is promoted to a global unknown. The far field enters at $-U$ through a Lagrange-multiplier condition. Only the upper half is solved. The midplane is a symmetry line. At low Oh a train of decaying capillary waves runs ahead of the rim. The sheet outline is a computed interface, $\mathrm{Oh}_{\mathrm{SB}} = 0.035$ at $t \approx 15$.

2 Non-dimensionalisation

Three conventions appear in this literature and they do not agree. Getting this wrong moves the answer by exactly $\sqrt{2}$, which is large enough to look like physics.

This code scales on the full thickness h_0, the density ρ and the surface tension σ, so that $\rho = \sigma = h_0 = 1$ and the half-thickness is 0.5.

quantity	scale
length	h_0
velocity	$\sqrt{\sigma/(\rho h_0)}$
time	$\sqrt{\rho h_0^3/\sigma}$
Ohnesorge	$\mathrm{Oh} = \mu/\sqrt{\rho\sigma h_0}$

Sanjay *et al.* [3] scale velocity on V_{TC} itself. Savva & Bush [7] use the same velocity scale and the same Ohnesorge definition as that paper, but a viscous clock $\tau_{\mathrm{vis}} = \mu H/(2\gamma)$ with H the full thickness, so $H \equiv h_0$. Sanjay *et al.* and Savva & Bush therefore share an identical Ohnesorge convention. This code differs from both by $\sqrt{2}$,

$$\mathrm{Oh}_{\mathrm{SB}} = \mathrm{Oh}_f = \mathrm{Oh}_{\mathrm{code}}/\sqrt{2}. \tag{2}$$

The coded $\mathrm{Oh} = 0.05$ used to match the Basilisk study is $\mathrm{Oh}_{\mathrm{SB}} = 0.035$. In code units, $\tau_{\mathrm{vis}} = \mu/2$. Throughout this document Oh means $\mathrm{Oh}_{\mathrm{SB}}$, because that is what the published comparison curves use. The velocity is reported as $u^* = U/V_{TC}$, which is directly comparable across all three conventions and tends to 1.

3 Numerical framework

Incompressible Navier–Stokes is solved in the liquid only,

$$\rho\big(\partial_t \mathbf{v} + \mathbf{v}\cdot\nabla\mathbf{v}\big) = -\nabla p + \nabla\cdot(2\mu\boldsymbol{\mathcal{D}}), \qquad \nabla\cdot\mathbf{v} = 0, \tag{3}$$

with the free surface carrying the stress condition $(-pI + 2\mu\boldsymbol{\mathcal{D}}) \cdot \mathbf{n} = \sigma\kappa\mathbf{n}$. There is no gas phase, hence no density ratio, no viscosity ratio, no diffuse interface and no spurious-current budget to defend. That is the principal methodological difference from the Basilisk study, which resolves both phases with a volume-of-fluid tracer and a Brackbill surface-tension force.

The retraction path is long. The tip travels $\sim V_{TC}\, t$, so a laboratory-frame domain must contain the entire excursion. The Basilisk study handled that with a domain $100\, h_0$ wide. Here the rim velocity is promoted to a single global degree of freedom $U(t)$, fixed by a Lagrange multiplier that pins the tip at the origin. The far field enters at $-U$. The mesh then only has to cover a window around the rim, and $U(t)$ is itself the physical observable.

The window must contain the viscous footprint $\sqrt{4\nu t}$, which grows with μ, so the required window grows with Oh. The campaign is bounded at $\mathrm{Oh}_{\mathrm{SB}} \leq 1$. Beyond that the cost grows sharply for the strongly viscous limit where no rim forms and the curves collapse. $\mathrm{Oh}_{\mathrm{SB}} \sim 1$ is where inertia and viscosity are comparable.

The solver uses BDF2 with adaptive time stepping. The output cadence sets the timestep, because maxstep is clamped to the current output interval and the temporal-error estimator never binds. The number of output points is therefore a physics-relevant parameter, exposed as `--n-outputs`. Section 6.3 shows that the campaign values are not sensitive to it at any level that matters.

4 Phenomenology

Figure 2 is the sequence. At $\mathrm{Oh}_{\mathrm{SB}} = 0.035$ a nearly circular rim is connected to the film by a distinct neck, with a train of decaying capillary waves running ahead of it, as Savva & Bush [7] and Brenner & Gueyffier [8] describe for $\mathrm{Oh} < 0.1$. At $\mathrm{Oh}_{\mathrm{SB}} = 0.2377$ the rim is still compact but the waves have gone. At $\mathrm{Oh}_{\mathrm{SB}} = 0.61945$ the neck has washed out. At $\mathrm{Oh}_{\mathrm{SB}} = 1$ there is no distinct rim. The sheet thickens over a broad region. The rim height at fixed t is remarkably insensitive to Oh, because the mass collected is set by the retraction velocity, which is itself Oh-independent. What changes is how that mass is distributed.

Figure 3 is the observable the campaign exists to measure. Every case is still climbing at the end of its production record. None reached the $u^* = 0.995$ target the sweep was sized for. The long inertial runs later closed that approach against V_{TC} itself. Figure 4 shows why a single global fit of the deficit is not usable if the record is short. The exponent in $1 - u^* \propto (t^*)^a$ runs from -0.435 at $\mathrm{Oh}_{\mathrm{SB}} = 0.035$ to -0.588 at $\mathrm{Oh}_{\mathrm{SB}} = 1$ on those production windows. Longer records, reported in §7, put the middle of the sweep on $-1/2$.

5 The one-dimensional reduced model

Savva & Bush's analysis is a long-wave (Trouton) reduction of the same problem [9, 7]. An independent finite-element implementation of that model lives in the repository so that a disagreement with their published figures can be attributed to digitisation, model, or code. It reproduces the early-time asymptote $u^* \to \sqrt{t^*/\pi}$ and the exact tip growth $h = 1 + t/(2\mu)$. Richardson extrapolation at $\mathrm{Oh}_{\mathrm{SB}} = 1$ gives $u^*(30) \to 0.98073$ as $\mathrm{d}t \to 0$. Against 0.95026 from the two-dimensional finite-element method and 0.9542 from the digitised Savva & Bush curve, the converged one-dimensional value is 3.2% and 2.8% high. An earlier report of a clean three-way agreement was an artefact of an under-resolved timestep.

The bulk capillary term is switched off in that solver. Savva & Bush's tip closure already is the capillary pull of the massless cap, so a bulk curvature term double-counts it. That is a physical argument supported by agreement, not a derivation. The detailed Trouton/Ahsan campaign, including an independent transcription of the code released with Ahsan *et al.* [10], is the viscous-limit note.

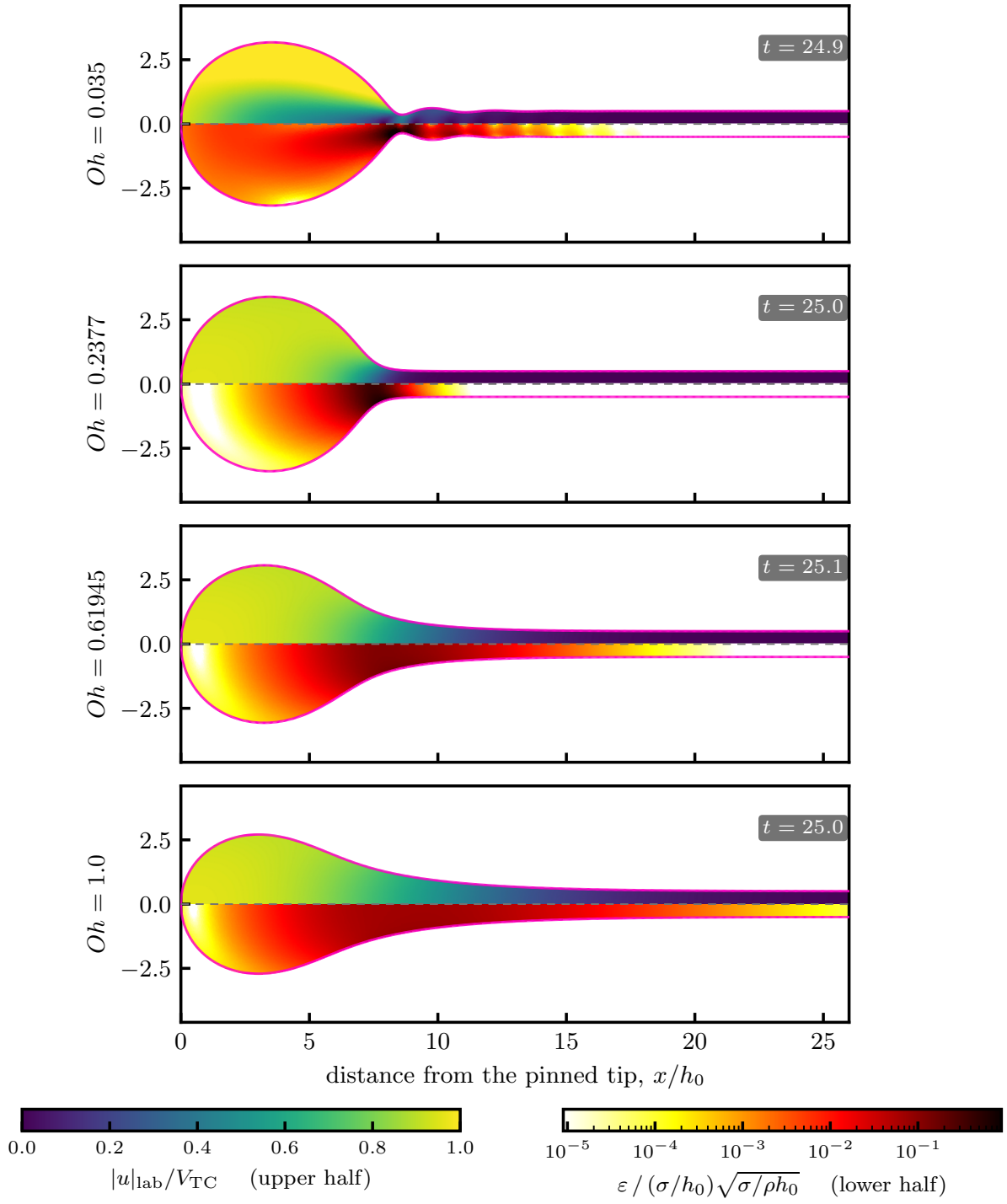


Figure 2: Velocity magnitude (upper half) and viscous dissipation rate $\varepsilon = 2\mu\,\boldsymbol{\mathcal{D}} : \boldsymbol{\mathcal{D}}$ (lower half) at a common inertio-capillary time $t \approx 25$, over a factor-29 range in Oh. Distances are from the pinned tip in units of h_0. Velocity is in the laboratory frame, so the undisturbed film reads zero. Colour scales are shared across the panels.

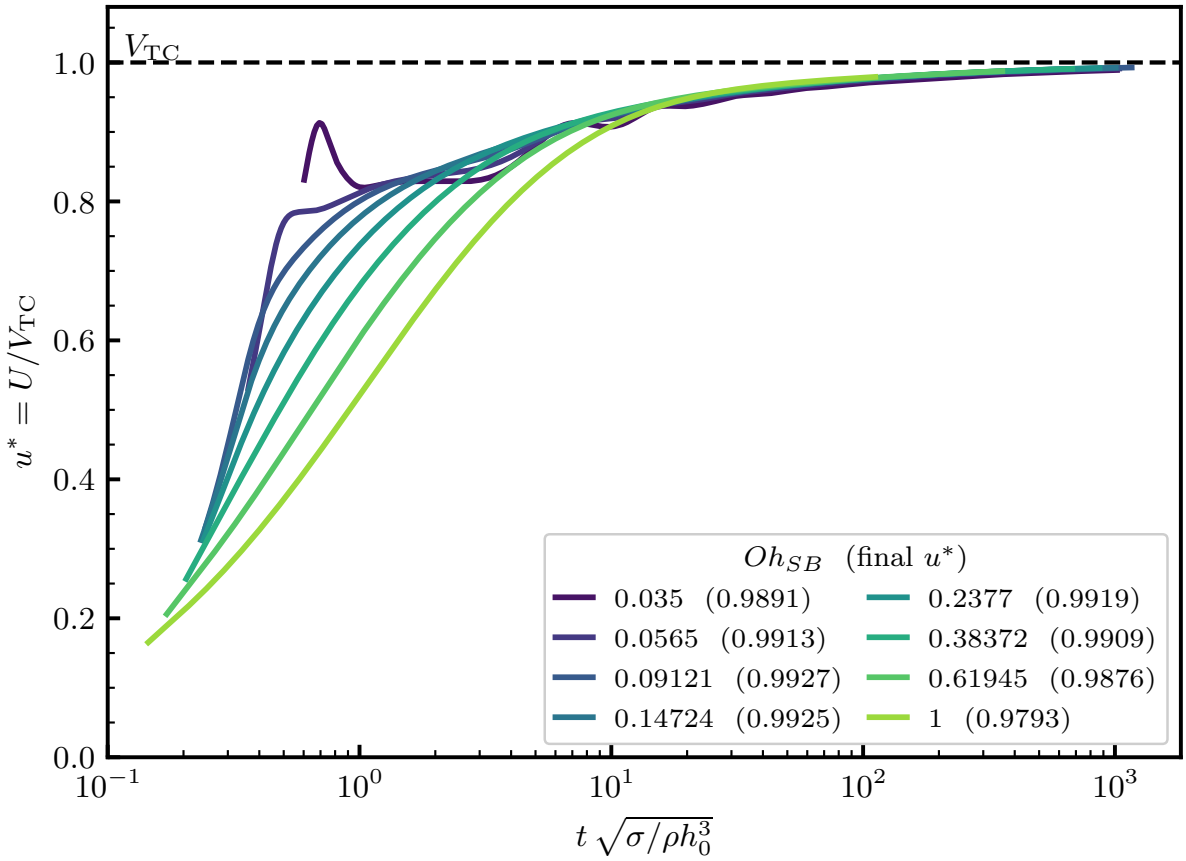


Figure 3: Recovered control velocity $u^* = U/V_{TC}$ against the inertio-capillary clock for the eight-point sweep, $\mathrm{Oh_{SB}} = 0.035$ to 1. The time axis is logarithmic because the solver writes on a geometric output schedule. Two features are physical. The low-Oh overshoot near $t \approx 0.7$, where the rim first forms, and the late-time ripples at Oh ≤ 0.1, which are the capillary-wave train beating against the rim.

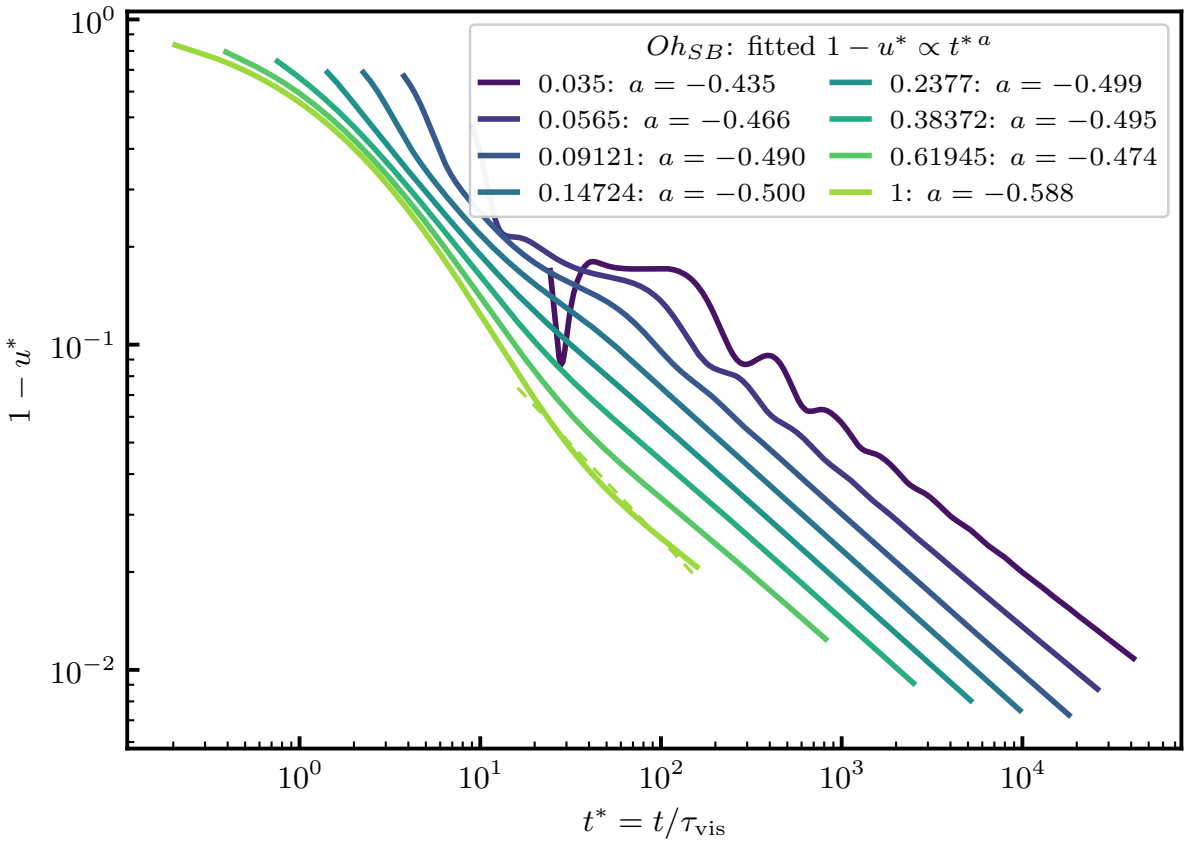


Figure 4: The same data as the gap $1-u^*$ against the viscous clock $t^* = t/\tau_{\mathrm{vis}}$, on log–log axes. Dashed lines are least-squares fits over the last decade of each record.

6 Verification

6.1 Window, not mesh

At $\mathrm{Oh}_{\mathrm{SB}} = 1$, $t^* = 30$, mesh resolution 0.03/0.04/0.06 and temporal tolerance agree to better than 0.01%. At window length $L = 12$ that agreement is around $u^* = 0.8191$, which is domain-limited.

L	$u^*(t^* = 30)$
12	0.81910–0.81915
16	0.91443
25	0.94942
40	0.95026
60	0.95026

The window length, not the mesh, is the controlling parameter, and it is converged for $L \geq 40$. A well-converged mesh on too small a window is the single easiest way to get a confidently wrong answer on this problem.

6.2 Savva & Bush

Converged $u^*(t^* = 30) = 0.95026$ at $\mathrm{Oh}_{\mathrm{SB}} = 1$ sits 0.41% below the digitised Savva & Bush figure-6 value of 0.9542. At $\mathrm{Oh}_{\mathrm{SB}} = 0.5$ the code gives 0.9297 against their ≈ 0.93. The 0.41% offset is not a timestep, mesh, or domain artefact. Whether it is digitisation error, a long-wave modelling error, or a genuine difference between the lubrication model and Navier–Stokes is the question flagged in §5.

6.3 Timestep

Because dt is set by the output cadence, the campaign values were checked against it at $\mathrm{Oh}_{\mathrm{SB}} = 1$, $t^* = 30$, $L = 40$.

`n-outputs`	$u^*(30)$	difference
100	0.9503405929	
200	0.9502772866	-6.331×10^{-5}
400	0.9502600436	-1.724×10^{-5}
800	0.9502529574	-7.086×10^{-6}

The total spread from the coarsest schedule tested to the finest is 8.8×10^{-5}, against the 0.41% (3.9×10^{-3}) offset from Savva & Bush that the campaign is trying to resolve. The campaign u^* values are not schedule-dependent at any level that matters. Runs may therefore use a coarse output cadence without loss of accuracy.

6.4 One-dimensional versus two-dimensional

The reduced model removes Oh from the problem by measuring length and time on the visco-inertial scales of Ahsan *et al.* [10]. Once x is measured in units of that length, the governing equations contain no parameter at all. The two-dimensional Navier–Stokes solution retains Oh. The two cannot agree across Oh in principle. The one-dimensional deficit falls as $(t^*)^{-2}$. The two-dimensional deficit falls as $(t^*)^{-1/2}$.

Dropping inertia outright is not the viscous-limit reduction. An inertialess balance forces a spatially uniform viscous stress, which no initially flat sheet can satisfy. Inertia is what localises the disturbance near the edge. What large Oh buys is that capillary stress leaves the bulk and acts only through the edge condition.

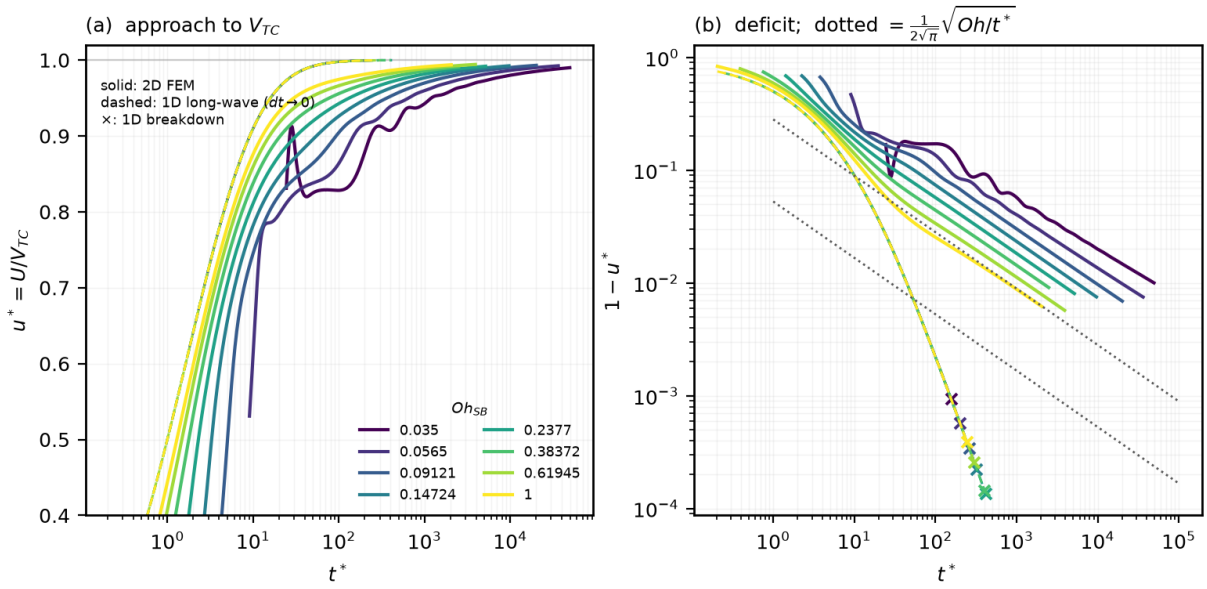


Figure 5: One-dimensional long-wave model against the two-dimensional finite-element solution. The reduced model is universal in t^*. The two-dimensional solution is not.

Compared at matched T against the converged reduced model, $\mathrm{Oh}_p = 10$ tracks it closely at first and then leaves it. Taking the slope gap exceeding 0.10 as the operational boundary gives a window

$$T_{\mathrm{dep}} \approx 0.84\,\mathrm{Oh}_p^{0.48} \tag{4}$$

over a factor 13 in Oh_p. The window widens as $\sqrt{\mathrm{Oh}_p}$, not as Oh_p. That is not what the stated validity bound $T \ll \mathrm{Oh}_p$ would lead one to expect. Extrapolating the fit, a decade of long-wave validity ($T_{\mathrm{dep}} = 10$) requires $\mathrm{Oh}_p \approx 170$, i.e. $\mathrm{Oh}_{\mathrm{SB}} \approx 85$. That is a four-point extrapolation over one decade of Oh and should be treated as an order of magnitude. The case that would test it is $\mathrm{Oh}_{\mathrm{SB}} = 50$, which is not yet run.

7 What is and is not established

Established. The co-moving formulation, and that the window (not the mesh) controls the answer, converged for $L \geq 40$ at $\mathrm{Oh}_{\mathrm{SB}} = 1$. $u^*(t^* = 30) = 0.95026$ at $\mathrm{Oh}_{\mathrm{SB}} = 1$, converged in mesh, window and timestep. A 0.41% offset from the digitised Savva & Bush value that is not numerical. Temporal error below 10^{-4} across an eightfold change in output cadence. The phenomenological sequence of figure 2 across a factor 29 in Oh.

The long-time limit. $v(t \to \infty) = V_{TC}$ at every Oh, approached diffusively as

$$1 - u^*(t^*) = \frac{1}{2\sqrt{\pi}}\sqrt{\mathrm{Oh}_{\mathrm{SB}}/t^*}. \tag{5}$$

Five cases spanning a factor 6.8 in Oh sit within 1% of the constant $1/(2\sqrt{\pi}) = 0.28209$, namely 0.2837, 0.2823, 0.2816, 0.2813, 0.2793 at $\mathrm{Oh}_{\mathrm{SB}} = 0.14724$, 0.2377, 0.38372, 0.61945, 1.0, with measured exponents -0.4901 to -0.4998 against the predicted $-1/2$. Viscosity sets how slowly V_{TC} is reached, not what is reached.

The coefficient is established over that band and not beyond it. Outside $0.14724 \leq \mathrm{Oh}_{\mathrm{SB}} \leq 1$ it departs in both directions. +44% at $\mathrm{Oh}_{\mathrm{SB}} = 0.035$, and a factor of seven low at $\mathrm{Oh}_{\mathrm{SB}} = 5$, where the exponent nonetheless returns to $-1/2$. The exponent is the robust feature. The coefficient is not universal on the evidence available.

The inertial planar finite-element versus Basilisk comparison at $\mathrm{Oh_{SB}} < 0.1$. Terminal finite-element u^* is 0.994–0.996. Basilisk at $\mathrm{Oh_{SB}} = 0.035$, taken to capillary time $t = 500$, is $u^* \approx 0.977$. The offset is the expected difference between a sharp free surface with no gas and a diffuse two-phase interface. This is a statement about V_{TC}, not about the coefficient in (5) below $\mathrm{Oh_{SB}} = 0.14724$.

The one-dimensional long-wave model is universal in t^* and the two-dimensional solution is not, so the two cannot agree across Oh in principle.

Not established. What happens to the deficit coefficient at low Oh. The three lowest production cases sit outside (5), and extending two of them moved them further from $1/(2\sqrt{\pi})$, not toward it. At $\mathrm{Oh_{SB}} = 0.035$ the deficit is not monotone, so no decaying power law can describe it. That oscillation is rim dynamics at the inertial end of the range. Pierson *et al.* [11] establish ringing about V_{TC} for an axisymmetric filament at $\mathrm{Oh} \lesssim 0.1$. Nobody states it for the planar sheet. Whether (5) is exact. Five cases agreeing to within 1% is suggestive of a closed form, not proof of one. How large Oh must be for long-wave asymptotics in two dimensions. The $\sqrt{\mathrm{Oh}_p}$ window is a result. The inference that a decade of validity needs $\mathrm{Oh}_p \approx 170$ is an extrapolation. Anything at $\mathrm{Oh_{SB}} > 1$, which is out of scope for the production campaign. The single diagnostic case at $\mathrm{Oh_{SB}} = 5$ is reported only where §6.4 needs it.

8 Reproducing the figures

Simulation output does not live in this repository. Runs write to the project run root on the compute host. Custody is recorded in the project tracker. The figure script takes paths to those run roots and skips any case whose data is not mounted. From the repository root,

```
uv run python postProcess/make_docs_figures.py \
    --out docs/planar-campaign/figures
```

The four campaign figures rebuild from that output. Figure 5 is produced by `postProcess/compare_1d_2d.py`.

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